

Rachit Sabharwal

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Professional Summary

Ph.D. Candidate (expected Aug 2026) with training in causal inference and quasi-experimental methods, plus 6+ years building predictive models and decision-support tools for business and research stakeholders across CPG, chemicals, biotech, and healthcare. Comfortable owning work end-to-end — from large-scale ETL and modeling through stakeholder-facing dashboards and executive white papers.

Education

- Ph.D. The University of Texas Health Science Center at Houston**, Epidemiology and Biostatistics Houston, TX
Expected Aug 2026
- Coursework: Causal Inference, Quasi-Experimental Methods, Survival Analysis, Longitudinal Data Analysis, Bayesian Statistics
 - Certificates: Advanced Data Science
- Cert. The University of California, Berkeley**, Software Development and Programming Berkeley, CA
Apr 2025
- M.S. The University of Texas Health Science Center at Houston**, Biostatistics Houston, TX
May 2022
- B.S. University of Rochester**, Environmental Health Rochester, NY
May 2018

Experience

- UTHealth Houston, School of Public Health**, Research Data Scientist Houston, TX
Feb 2020 – present
- Contributed quantitative analyses on claims data and EHR data to an R01 grant proposal on disability in adolescents and young adults.
 - Cut data-munging turnaround 24x by re-architecting the TexasCARES ETL to process multi-year, out-of-memory cohort data using polars and duckdb; stood up CI/CD automated reporting that ingests new data weekly, runs automated data-quality checks, and publishes stakeholder outputs without manual intervention.
 - Shipped and maintained a public-facing statewide COVID-19 dashboard (Python, Tableau) ingesting case and antibody survey data continuously, providing real-time epidemic monitoring for public and health-department stakeholders.
 - Designed multi-site star schema and ETL for DETECT-RPC harmonizing clinical records from 6 academic medical centers; built a Quarto dashboard giving PIs real-time visibility into data collection progress across sites.
 - Co-authored 4 peer-reviewed publications on SARS-CoV-2 vaccine non-response and pediatric COVID comorbidities using survival analysis and mixed models on population-based cohorts.
- The Dow Chemical Company**, ML R&D Intern Lake Jackson, TX
May 2025 – Aug 2025
- Benchmarked Bayesian models (Gaussian process regression, Bayesian neural networks, Monte Carlo dropout) against frequentist counterparts across 25+ measurement-error scenarios using PICP, ECE, and CI-width calibration metrics.
 - Shipped an R Shiny decision-support app automating 50% of enterprise Life Cycle Assessment workflows, estimated at \$15M in annual cost savings across Dow's global operations.
 - Authored an internal Bayesian UQ white paper distilling benchmark findings into organizational guidance for future predictive modeling projects.
- Genentech**, Informatics Intern San Francisco, CA
May 2021 – Jan 2022
- Reproduced the published GLIT multimodal GNN on DILI benchmarks, then prototyped a lower-complexity MLP combining ECFP fingerprints with drug diffusion profiles, demonstrating comparable predictive performance at substantially reduced model complexity.
 - Built a document tokenization framework that feeds into downstream NLP annotation and text-mining pipelines.

Procter & Gamble , Advanced Analytics Co-Op	Cincinnati, OH
- Delivered predictive and explanatory models on shopper and retailer behavior for global Fabric Care brand teams; translated findings into recommendations that informed corporate strategy around DTC ecommerce. - Raised cross-functional analyst throughput 1.5x by building GCP big-data ETL pipelines using Apache Airflow to process out-of-memory shopper and retailer datasets. - Raised codebase test coverage from 0% to 70% by introducing containerization, unit testing, and agile practices.	Jan 2021 – Apr 2021
Bristol Myers Squibb , Data Engineering Intern	San Francisco, CA
- Built PatRec, a full-stack patent recommendations app chaining chemical-entity NER, NCBI federated search, and TF-IDF cosine-similarity ranking to surface relevant patents from an 80M-patent corpus for internal R&D scientists. - Designed relational (PostgreSQL) and graph (Neo4j) database architectures supporting multiple internal R&D analytical workloads.	June 2020 – Aug 2020

Selected Publications

Trust and Uncertainty Quantification in Machine Learning Models Under Measurement Error Sabharwal R (The Dow Chemical Company, Internal White Paper)	Aug 2025
Factors associated with elevated SARS-CoV-2 immune response in children and adolescents Messiah SE, Abbas R, Bergqvist E, Swartz MD, Talebi Y, Sabharwal R , et al. 10.3389/fped.2024.1393321 (Frontiers in Pediatrics)	Aug 2024
Baseline characteristics of SARS-CoV-2 vaccine non-responders in a large population-based sample Yaseen A, DeSantis SM, Sabharwal R , et al. 10.1371/journal.pone.0303420 (PLoS One)	May 2024
Long-term immune response to SARS-CoV-2 infection and vaccination in children and adolescents Messiah SE, Talebi Y, Swartz MD, Sabharwal R , et al. 10.1038/s41390-023-02857-y (Pediatric Research)	Oct 2023

Skills

Causal Inference & Experimentation: Difference-in-Differences, Synthetic Control, Propensity Scores / IPW, G-methods, Instrumental Variables, Target Trial Emulation, A/B Testing & Power Analysis

Statistical Methods: Survival Analysis, Linear & Generalized Linear Mixed Models, Longitudinal Data Analysis, Bayesian Inference & UQ, Measurement Error Modeling

ML: scikit-learn, PyTorch, tidymodels, brms, statsmodels, lifelines, Optuna, SHAP

Programming & Data: Python (polars, pandas, duckdb), R (data.table, tidyverse), SQL, SAS, Bash

Production & Cloud: Docker, Apache Airflow, GitHub/GitLab CI/CD, GCP (BigQuery), AWS, Azure, PostgreSQL

Visualization & Communication: R Shiny, Quarto, Tableau, ggplot2, Plotly, Streamlit, peer-reviewed publications, white papers